

Spinello & Stilwell state-dependent Gaussian observation noise estimation.

31/08/26

Clearing up convention & Glossary first

- The paper writes gradients as row vectors. So $h' = \nabla h$ is a row, and $h'^T h$ is column-time-row, an $n \times n$ matrix.

Paper	What it is	Mine
x	latent state	x , same
z	measurement	my y
$h(x)$	observation mean map	observation map ('observation' in Linear Gaussian Models)
$\Sigma(x)$, scalar $\sigma(x)$	state-dep obs noise variance	$R(x)$
$\tilde{z} = z - h$	innovation	same
$\hat{x}[k k-1]$, $P[k k-1]$	predicted mean and covariance	prior mean μ & prior spread σ^2
$\hat{x}[k k]$, $P[k k]$	updated mean and covariance	posterior mean & spread of q
$\ell(x)$	negative log-likelihood objective	clash but related: The paper's ℓ is the whole negative-log objective. My ℓ is the c_2/c_4 term is $\log R$ at the prior mean. Different objects. Connected via the paper's troublesome term is $\frac{1}{2} \ln \sigma$, which is $\frac{1}{2} \ell$ in my notation. The object my entire gap expansion is built from is the object causing their pathology
$r(x)$	stacked residual vector	internal to Gauss-Newton, no spread
$s(35c)$	gradient of the measurement, port	exact
$P(35d)$	Gauss-Newton curvature matrix	The paper's R is the curvature matrix, the $J^T J$. Nothing to do with noise. The notation has consistently snapped places between literatures
$\bar{U}(35e)$	Fisher information from the measurement	sets posterior precision, like my Gaussian recursive precision update
$\frac{1}{2} \ln \sigma$	the normaliser penalty	$\frac{1}{2} \ell$ in my $\ell = \log R$ notation
Iterated to convergence	iterated EKF	the Spinello-Stilwell rung of my reference ladder

01/09/2026

What the p^* programme ladder rung has to produce

The ladder reports.

$$\text{gap} = E_{y \sim p} [KL(q(x) \parallel p(x|y))]$$

$p(x|y)$ is the exact posterior. In cpomdp the S2 Lattice computes $p(x|y)$.

$q(x)$ is whatever the rung hands back. For KL to be the thing the ladder measures, q must be a proper distribution over x , for this rung it is Gaussian.

So, the rung's job per timestep: take a Gaussian prior $N(\mu, P)$ and one observation y , and return a Gaussian $q = N(\hat{x}, P_*)$. Two numbers, the mean & covariance.

So the mean is needed to be a MAP estimate. The best single point to centre a Gaussian on is the posterior mode and to find the mode that means minimising the negative log posterior, turning an inference problem into an optimisation problem. This is where Gauss-Newton is used.

And the covariance is a Fisher information. A Gaussian centred on the mode with covariance equal to inverse curvature at the mode is the Laplace approximation. The Stillwell paper uses the expected curvature, the Fisher information, rather than the observed one.

Step 1: the objective

The posterior is prior times likelihood:

$$p(x|y) \propto N(x; \mu, P) \cdot N(y; h(x), \sigma(x))$$

From the paper take minus the log. Drop constants that don't depend on x .

$$\begin{aligned} \ell(x) &= \frac{1}{2} \xi^T P^{-1} \xi & \xi &= x - \mu \\ &+ \frac{1}{2} z^2 / \sigma(x) & z &= y - h(x) \\ &+ \frac{1}{2} \ln \sigma(x) \end{aligned}$$

Term 1: distance from the prior mean, weighted by prior precision.

Term 2: measurement mismatch, weighted by noise precision

Term 3: the gaussian normalisation

The reason term 3 exists is that a Gaussian density carries $1/\sqrt{\sigma}$ in front. When σ is a constant, $\ln \sigma$ is constant & drops. With our $R(x)$ the normaliser is a function of the thing we're minimising over so it stays.

In cpondp notation it is $\frac{1}{2} L$ with $L = \log R$. Meaning, initially, the object our gap expansion is built from is the arithmetic obj, causing trouble here!

Note: Why this is the right objection for cpondp. It is exactly minus the log of the posterior the lattice computer. Same prior, same likelihood, same $R(x)$. The runy and the ground truth are minimising and integrating the same function respectively. No approximations - yet

step 2: The residual vector / Gauss - Newton justification

Gauss - Newton needs $L = \frac{1}{2} r^T r$ so we write each, penalty as half a square:

$$r(x) = \left(\underbrace{P^{-1/2} \xi}_r, \underbrace{\sigma^{-1/2} z}_r, \underbrace{(\ln \sigma)^{1/2}}_{r_3} \right)^T$$

r_1 & r_2 are honest real residuals. r_3 is real only when $\sigma > 1$. Below that, $\ln \sigma < 0$, r_3 is imaginary. The paper's footnote does concede this and note the 'update equation' stay real. This is true and not the problem we will run into.

side check

$\frac{1}{2} (r_1^2 + r_2^2 + r_3^2)$ gives the $L(x)$ equation back exactly

step 3: The Gauss - Newton step

$$\hat{x}^{(L+1)} = \hat{x}^{(L)} - [\nabla^T r \nabla r]^{-1} \nabla^T r$$

Three reasons I chose this as the correct method and not just an available one.

- 1) It is the iterated EKF. Bell and Calvey 1993: one step of the above from $\hat{x}^{(0)} = \mu$ is the EKF update, algebraically. Iterating to convergence is the iterated EKF. Meaning the runy is the standard Gaussian filter for this likelihood. Not bogus.
- 2) It carries a shape guarantee. For real J , $w^T (J^T J) w = \|Jw\|^2 \geq 0$. The step matrix is PSD. Every step is a descent direction. Full Newton, with its $\sum r_i \nabla^2 r_i$ term, has no shape guarantee and needs second derivatives of $\sigma(x)$.
- 3) The prior term is free damping. The step matrix splits as $P^{-1} + R$ where P^{-1} comes from r_1 and is positive definite on its own. So $P^{-1} + R \geq 0$ wherever $R \geq 0$. That is what a different method Levenberg - Marquardt gives for with a λI . Our Bayesian prior gives it away.

Specifically for cpondp $\Rightarrow$ Our observation map is linear. $\nabla^2 h = 0$. So the second derivative term of Gauss - Newton discards contains only $\nabla^2 \sigma$ contributions. The approximation is strictly smaller than the on the paper made for arctangent bearings.

Step 4 The three matrices (35c-e) - derived from r

Now we differentiate r in the scalar case. Row vectors throughout.

$$\begin{cases} \partial r_1 / \partial x = p^{-1/2} \\ \partial r_2 / \partial x = -\sigma^{-1/2} (h' + (\zeta/2\sigma)\sigma') \\ \partial r_3 / \partial x = \frac{1}{2} (\ln \sigma)^{-1/2} \sigma'/\sigma \end{cases}$$

Gradient

$$\begin{aligned} J^T r = g &= P^{-1} \xi + r_2 \partial r_2 + r_3 \partial r_3 \\ &\Rightarrow P^{-1} \xi - (\zeta/\sigma) h' - (\zeta^2/2\sigma^2) \sigma' + (1/2\sigma) \sigma' \\ &\Rightarrow P^{-1} \xi + \underbrace{\left[-(\zeta/\sigma) h' + (1/2\sigma) (1 - \zeta^2/\sigma) \sigma' \right]}_{\text{S, eq. 35c from paper}} \end{aligned}$$

Side note on the r_3 term: $(\ln \sigma)^{1/2} \cdot (\frac{1}{2} (\ln \sigma)^{-1/2} \cdot \sigma'/\sigma) = \sigma'/2\sigma$

Logs cancel. Real for any $\sigma > 0$ & sign of $\ln \sigma$ is irrelevant.

Curvature

$$\begin{aligned} J^T J &= P^{-1} + (\partial r_2)^T (\partial r_2) + (\partial r_3)^T (\partial r_3) \\ &\Rightarrow P^{-1} + \left(\frac{1}{\sigma} \right) (h' + (\zeta/2\sigma) \sigma')^T (h' + (\zeta/2\sigma) \sigma') \\ &\quad + \underbrace{\sigma'^T \sigma' / (4\sigma^2 \ln \sigma)}_{\text{R, eq. 35d in paper}} \end{aligned}$$

R, eq. 35d in paper

Side note: Nothing cancels in the r_3 piece this time.

The r_2 line closer

$$r_2(x) = \sigma(x)^{-1/2} \cdot \zeta(x) \text{ where } \zeta = y + h$$

Product rule

For any power n the derivative of u^n is $n u^{n-1} u'$.
here $n = -1/2$

$$\zeta = y + h(x)$$

$$\zeta' = 0 + h' = h'$$

$$\frac{d}{du} [u^{-1/2}] = -\frac{1}{2} u^{-3/2}$$

$$\frac{d}{dx} [\sigma^{-1/2}] = -\frac{1}{2} \sigma^{-3/2} \cdot \sigma'$$

$$\begin{aligned} u &= \sigma^{-1/2} & u' &= -\frac{1}{2} \sigma^{-3/2} \\ v &= \zeta & v' &= h' \end{aligned}$$

$$\frac{\partial r_2}{\partial x} \Rightarrow -\frac{1}{2} \sigma^{-3/2} \sigma' \cdot \zeta + \sigma^{-1/2} h'$$

$$\frac{\partial r_2}{\partial x} = -\frac{1}{2} \zeta \sigma^{-3/2} \sigma' - \sigma^{-1/2} h'$$

organise around $\sigma^{-1/2}$

$$\begin{aligned} \frac{\partial r_2}{\partial x} &= -\sigma^{-1/2} h' - \sigma^{-1/2} (\zeta/2\sigma) \sigma' \\ &\Rightarrow -\sigma^{-1/2} (h' + (\zeta/2\sigma) \sigma') \end{aligned}$$

Now the Fisher Information so I'll write each line because this is very new to me and had to learn as I went.

The object $\bar{U} = E[s^T s]$

s is the measurement part of the gradient, from step 4. The expectation is over the innovation z , treated as random with $z \sim N(0, \sigma)$

Worth stating explicitly because this took me a moment to get my head around. The reason 'random' is because $\bar{U}$'s job is different from R 's job. R shapes the step for this measurement, so it uses the actual z . $\bar{U}$ sets the reported posterior precision, which answers "how much does a measurement from this sensor, at this state, teach you on average."

Averaging over what the sensor could emit means averaging over z . Pick out the state in question, the model says $y = Hx$ plus noise, so $z = y - h$ is the noise itself: mean 0, variance σ .

The recursion this feeds is from the paper's 35b equation.

$$P_+^{-1} = P^{-1} + \bar{U}$$

In english this reads as Posterior precision equals prior precision plus information from the measurement. $\bar{U}$ is the "plus information" bit

The moments toolbox we need

4 facts about $z \sim N(0, \sigma)$

$E[z] = 0$ mean zero

$E[z^2] = \sigma$ the variance

$E[z^3] = 0$ odd moment of a symmetric distribution

$E[z^4] = 3\sigma^2$ the Gaussian fourth moment

We'll need these

Now two pieces of s

from 35c

$$s = \underbrace{-(z/\sigma)h'}_a + \underbrace{(\frac{1}{2}\sigma)(1 - z^2/\sigma)}_c \sigma'$$

a is the ordinary Kalman innovation term
 c is the new term the prior mode brings, both are row vectors

So $s = a + c$ with

$$a = -(z/\sigma)h' \quad c = (\frac{1}{2}\sigma)(1 - z^2/\sigma)\sigma'$$

Small sanity check, we know $E[s] = 0$ to be a score function

$$\text{so } E[a] = -(E[\zeta]/\sigma)h' = 0 \quad \checkmark$$

$$E[c] = (1/2\sigma)(1 - E[\zeta^2]/\sigma) \sigma' = (1/2\sigma)(1 - \sigma/\sigma) \sigma' = 1/2\sigma(1-1)\sigma' = 0$$

Now Square it.

$$s^T s = (a+c)^T (a+c)$$

$$= a^T a + a^T c + c^T a + c^T c$$

Take expectations one at a time

$$a^T a = (- (\zeta/\sigma) h')^T (- (\zeta/\sigma) h')$$

$$\Rightarrow (\zeta^2/\sigma^2) h'^T h$$

$$\text{Apply } E[\zeta^2] = \sigma$$

$$E[a^T a] = (\sigma/\sigma^2) h'^T h = (1/\sigma) h'^T h$$

First term of SSE done $\checkmark$

Term 2

$$a^T c = - (\zeta/\sigma) \cdot (1/2\sigma)(1 - \zeta^2/\sigma) \cdot h'^T \sigma'$$

$$\text{scalar part} = -(1/2\sigma^2)(\zeta - \zeta^3/\sigma)$$

Every power of ζ in $\uparrow$ is odd so using the toolbox

$$E[\zeta - \zeta^3/\sigma] = 0 - 0 = 0$$

$$\text{so } E[a^T c] = 0$$

b. $c^T a$ is the same with the matrices in the other order so 0.

So in the printed R, the cross terms are alive because they are the actual innovations. On average they contribute nothing because the observation channel and the noise channel are uncorrelated sources of information.

Term 4

$$c^T c = (1/4\sigma^2) (1 - \zeta^2/\sigma)^2 \sigma'^T \sigma' \quad | \quad (1 - \zeta^2/\sigma)^2 = 1 - 2\zeta^2/\sigma + \zeta^4/\sigma^2$$

$$E[(1 - \frac{3}{4}\sigma^2)^2] = E[1] - E[\frac{3}{2}\sigma^2] + E[\frac{9}{4}\sigma^4]$$

$$\Rightarrow 1 - 2 + 3$$

$$= 2$$

sub back in

$$E(c^T c) = (2/4 \sigma^2) \sigma'^T \sigma' = (1/2 \sigma^2) \sigma'^T \sigma'$$

Second term of 35e done

$$\bar{U} = 1/\sigma h'^T h' + 0 + 0 + (1/2 \sigma^2) \sigma'^T \sigma'$$

$$\Rightarrow 1/\sigma h'^T h' + (1/2 \sigma^2) \sigma'^T \sigma'$$

This shows the modification leaves 35e rebated

$\bar{U} \geq 0$ unconditionally. Two outer products, coefficients $1/\sigma > 0$ and $1/2 \sigma^2 > 0$. so ~~$P_+^{-1} = P^{-1} + \bar{U} \geq 0$~~

$$P_+^{-1} = P^{-1} + \bar{U} \geq 0$$

Expected versus observed curvature. If we compare 35e to 35d term for term the averaging did three things

- ① killed the gross terms (odd in ζ)
- ② Replaced the ζ^2 -dependent coefficient on $\sigma'^T \sigma'$ with its mean
- ③ produced no counterpart to the $1/\ln \sigma$ term.

P is per measurement and can be a ~~scalar~~ matrix. $\bar{U}$ is the average and ~~scalar~~ is not.

Another quick check (Note to self *: make a python test)

set $\sigma' = 0$ in the result & you get $\bar{U} = 1/\sigma h'^T h'$ which is the standard Kalman information for a linear observation with final noise.

4/09/26

Step 5

Compare the expansion of the r_2 row against the printed 35d

$$\left(\frac{1}{\sigma}\right)h' + \left(\sqrt{2}\sigma\right)o')^T(h' + (\sqrt{2}\sigma)o') = \left(\frac{1}{\sigma}\right)h'^T h' + (\sqrt{2}\sigma^2)(h'^T o' + o'^T h' + (\sqrt{2}^2/4\sigma^3) o'^T o'$$

These are the first 3 terms of 35d term for term 50

$$R = (\partial r_2)^T (\partial r_2) + (\partial r_3)^T (\partial r_3)$$

Step 6

First a refresher on the damping argument from step 3

The step matrix in the update (35a) is not R alone. It is $M = P^{-1} + R$

The P^{-1} is there because the residual vector had three rows with the first row being $r_1 = P^{-1/2} \xi$, which contributes its own block to

$$(\partial r_1)^T (\partial r_1) = (P^{-1/2})^T (P^{-1/2}) = P^{-1}$$

Now by logic P is a covariance so P^{-1} is symmetric positive-def. Strictly! Every direction w gives $w^T P^{-1} w > 0$.

If we add any PSD matrix to it

$$w^T (P^{-1} + R) w = w^T P^{-1} w + w^T R w > 0 + 0$$

Strictly positive for every w . So $M > 0$ meaning M is invertible and $-M^{-1}g$ pointed downhill. The prior term acts as a floor.

Levenberg-Marquardt ~~had~~ builds this floor by hand, adding λI with a tuning knob. Here we have a natural floor of P^{-1} coming from the Bayes prior.

We only need $w^T R w \geq 0$ for this argument to hold so

From steps 4 & 5, the curvature is exactly two pieces, one per Jacobian row:

$$R = (\partial r_2)^T (\partial r_2) + (\partial r_3)^T (\partial r_3)$$

Interrogate each separately. Pick an arbitrary direction w and ask about the sign of $w^T(\text{piece})w$.

r_2 block

from the r_2 steps

$$\partial r_2 / \partial \sigma = -\sigma^{-1/2} b$$

$$\text{where } b = h' + (\zeta/2\sigma)\sigma'$$

$$w^T (\partial r_2)^T (\partial r_2) w$$

$\partial r_2 \cdot w$ is a row times column so a scalar. Call it t .

$$t = \partial r_2 \cdot w = -\sigma^{-1/2} \cdot b \cdot w$$

then becomes $t \cdot t = t^2$

The three ingredients of t are real: $\sigma^{-1/2}$ is real because $\sigma > 0$ (variance), b is real because h', ζ, σ' are all real quantities of the model, w is real because we set it so.

$t^2 \geq 0$ for every w , every ζ , every state, every $\sigma > 0$.

$\therefore$ This piece is PSD everywhere the mode is defined.

r_3 block

from step 4:

$$\frac{\partial r_3}{\partial x} = \frac{1}{2} (\ln \sigma)^{-1/2} \cdot \sigma' / \sigma$$

surround the same way, $(\partial r_3)w$ is scalar again

$$u = \frac{1}{2} (\ln \sigma)^{-1/2} \frac{\sigma'}{\sigma} \cdot w$$

$$u^2 = \frac{1}{2} \cdot \ln \sigma^{-1} \cdot \frac{(\sigma' w)^2}{\sigma^2} \Rightarrow \frac{(\sigma' w)^2}{4 \sigma^2 \ln \sigma}$$

Inspecting the 3 terms
 $(\sigma' w)^2 \geq 0$ a real square
 $4 \sigma^2 > 0$ a positive square
 $\ln \sigma$? sign depends on σ

$\sigma > 1$: $\ln \sigma > 0$ piece > 0
 $\sigma = 1$: $\ln \sigma = 0$ pole: division by 0
 $\sigma < 1$: $\ln \sigma < 0$ piece < 0 , strict negative when $\sigma' w \neq 0$

so the anatomy of R is now fully:

$R = \{ \text{piece that is (real)}^2, \text{PSD unconditionally} \}$
 $+ \{ \text{piece whose sign tracks } \ln \sigma, \text{negative exactly where } \sigma < 1 \}$

We can delete the bad term as covered in the paper with justification

$$R_{\text{mod}} = (\partial r_2)^T (\partial r_2) = \frac{1}{\sigma} b^T b$$

So the more: drop the $\frac{1}{4\sigma^2 \ln \sigma} \sigma^T \sigma$ block from R (35d).
Keep s (35c), $\bar{U}$ (35e), objective (18) rebutton

① Buys PSD outright. $R_{\text{mod}} = \frac{1}{\sigma} b^T b$, a real square, so $R_{\text{mod}} \geq 0$ everywhere. With $p^{-1} \geq 0$ as σ to floor: $M \geq 0$, every step descends.

No unit box or assumption on when iterates go.

② Gauss-Newton's PSD guarantee is a theorem about real Jacobians. $r_3 = (\ln \sigma)^{1/2}$ is imaginary for $\sigma < 1$ which is in the paper's own footnote.

③ Surgical $\rightarrow$ Fixed point lives in $g = 0$, and $\ln \sigma$ cancels in the gradient ($r_3, \partial r_3 = \sigma/2\sigma$, already in 35c). $\bar{U}$ never contains it.

④ Section 4 of the paper constants $\Rightarrow \sigma < 1$ throughout figs 2-4 for any $N \geq 3$, so the 1 term was negative in every published run and the filter won anyway.

Cost: no longer literal Gauss-Newton on 2σ . Name it in Paper 2!

* * We will test in python whether this change truly is surgical. Be interesting to see through others

Step 7 Invariant check

If we take a measurement and change its units by multiply all by the same constant $\lambda > 0$ what happens?

$$y \rightarrow \lambda y$$

Radians to degrees is $\lambda = 180/\pi$ so nothing physical happens only labels on numbers change

so what to check, every quantity the filter is built from should return unchanged. The filter's output is distribution over the state, and the state is never touched

The observation map, the model says $y \approx h(x)$. If y is now measured in different units, the prediction must be too

$$h \rightarrow \lambda h$$

The innovation

$$z = y - h \rightarrow \lambda y - \lambda h = \lambda z$$

Both terms scale so the difference scales.

The variance. Variance is an expectation of a square:

$$\sigma = E[(y - \lambda)^2] \Rightarrow E[(\lambda y - \lambda \lambda)^2] = \lambda^2 E[(y - \lambda)^2] = \lambda^2 \sigma$$

The λ comes out squared

The derivatives. Diff in x doesn't touch λ so they just inherit:

$$h' \rightarrow \lambda h' \quad \sigma' \rightarrow \lambda^2 \sigma'$$

The state side objects. x, μ, P, ξ, w remain the same because λ never reaches them.

so: $\boxed{h \rightarrow \lambda h \quad \xi \rightarrow \lambda \xi \quad \sigma \rightarrow \lambda^2 \sigma \quad h' \rightarrow \lambda h' \quad \sigma' \rightarrow \lambda^2 \sigma'}$

Count powers of λ . A term is invariant when the powers cancel to $\lambda^0 = 1$

The gradient $s = -\frac{1}{2} \left(\frac{\xi}{\sigma} \right) h' + \left(\frac{1}{2} \frac{\xi}{\sigma} \right) (1 - \frac{\xi}{\sigma}) \sigma'$

$\xi/\sigma h' \rightarrow \lambda \cdot \lambda / \lambda^2 = \lambda^0 \quad \checkmark$

the inner: $\xi/\sigma \rightarrow \lambda^2 / \lambda^2 = 1 = \lambda^0 \quad \checkmark$

outer: $\sigma'/\sigma \rightarrow \lambda^2 / \lambda^2 = \lambda^0 \quad \checkmark$

The curvature R : $R_{\text{mod}} = (1/\sigma) (h' + (\xi/2\sigma)\sigma')^T (h' + (\xi/2\sigma)\sigma')$ expanded into

$1/\sigma h'^T h \rightarrow \lambda \cdot \lambda / \lambda^2 = \lambda^0 \quad \checkmark$

$(\xi/2\sigma^2) (h'^T \sigma' + \sigma'^T h') \rightarrow \lambda \cdot \lambda \cdot \lambda^2 / \lambda^4 = \lambda^0 \quad \checkmark$

$(\xi^2/4\sigma^3) \sigma'^T \sigma' \rightarrow \lambda^2 \cdot \lambda^2 \cdot \lambda^2 \cdot \lambda^0 = \lambda^0 \quad \checkmark$

If we do the full R not R_{mod} the term is

$\sigma'^T \sigma' / 4\sigma^2 \ln \sigma$ numerator $\lambda^2 \cdot \lambda^2 = \lambda^4$ & $\sigma^2 = \lambda^4$ so they go

$\ln(\sigma) \rightarrow \ln \lambda^2 \sigma = \ln \sigma + 2 \ln \lambda$ antenard ~~so~~ so now

$\sigma'^T \sigma' / 4\sigma^2 \ln \sigma \rightarrow \sigma'^T \sigma' / 4\sigma^2 (\ln \sigma + 2 \ln \lambda)$ not invariant $\times$

The Fisher $\bar{U}$

$$1/2 h' T h' \rightarrow \lambda^0 \quad \checkmark$$

$$1/2 \sigma' \sigma' T \sigma' \rightarrow \lambda^1 \quad \checkmark$$

so the final tally is 7 out of 9. Even more evidence, with step 6's block to banish it into the shadow realm

Why the ladder is entitled to demand invariance

$$\text{gap} = E_{y \sim p^*} [KL(q(x) || p(x|y))] \quad \text{a number in nats}$$

- The KL compares two distributions over the state. Rescaling $y \rightarrow \lambda y$ only relabels the conditioning value. For the same physical measurements, q and $p(x|y)$ are unchanged, so the KL is unchanged.
- The outer expectation visits the same physical ~~deg~~ observations with the same probabilities.
- The gap is unit free by construction, before any fitter enters to the python lattice computes exactly that number.
- A single run is an approximation of that number. If the output moves with λ , it isn't a rough approximation, it's something else.
- With the $1/\ln \sigma$ term in the same physical problem in degrees vs radians $\rightarrow$ two different reported gaps for one facet.
- Step 6 faults the route (asent step possible), this faults the report / unit dependent number. 2 reasons same term convicted.
- Executable form: test $\rightarrow$ rescale y , rerun, gap must match to precision. Unmodified fails by the predicted $2 \ln 7$ shift, modified should pass.

Step 8 - The deletion is surgical logical justification points

- Fixed point of $x \leftarrow x - M^{-1}g$ is where $g = 0$. M steers the route & g picks the destination
- Gradient $r_s : dr_s = \sigma'/2\sigma$. The logs cancelled. Read everywhere already can be seen in the papers 35c. S verbatim
- Curvature $(\partial r_s)^T (\partial r_s) = \sigma'^T \sigma' / 4\sigma^2 \ln \sigma$. Nothing cancels but this is the deleted block.
- $\bar{U}$ built from s alone, no $\ln \sigma$ ever seen. 35e verbatim. P_+ untouched.

• Objective ~~is~~ untouched $\Rightarrow$ same mode

• Net: same $\bar{x}$, same P_+ , same g , same gap in nets. only the path of iterates changes. Provably descends

• The honest method is now a modified-metric Newton with the GN fixed point not a literal GN. Where $\sigma < 1$ we need a repair $\Rightarrow$ printed R claims to be $\nabla^T R \nabla R$ and every read one is PSD.

Step 9 the original paper never casted the term in.

This is referencing section 4 of the paper as N from [41]. This will be pinned in the python code

Step 10 - cpomdp's open communication/certifiability obligations

• Name: Spiniello - Stillwell iterated filter with a documented modification, will declare at first use in paper 2

• Three gtest test planned:

① Gap invariant under $y \rightarrow \lambda y$. fails unmodified ($2 \ln \lambda$ shift), passes modified. Blind to a wrong fixed point

② $\sigma' = 0 \Rightarrow$ exact agreement with Kalman but blind to all σ terms

③ $\sigma \gg 1 \Rightarrow$ near agreement with printed 35d. Catches any unexpected changes but blind to $\sigma < 1$.

• Logic for cite & repair over from scratch: published derivation exists, known EKF lineage, one declared change with a cost python can monitor & test.